Mark Lodewijk Helwig

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EDUCATION

University of Texas at Austin Graduating May 2027

B.S. Electrical and Computer Engineering Honors **GPA: 3.88**

- Relevant Courses: *Algorithms, Real-time Digital Signal Processing Lab, Digital Signal Processing Theory, Gateway to Robotics, Probability and Random Processes Honors*
- Current Courses: *Aerial Robotics, Robot Mechanism Design, Automatic Control*

WORK EXPERIENCE

Persona AI, *Locomotion Engineering Intern* (Pensacola, FL) May 2026 - Present

- Engineering behaviors across multiple humanoid hardware platforms using Nvidia's IsaacLab.
- Developing a pipeline for logging and processing hardware and simulation data.

Human Centered Robotics Group, *Researcher* (Austin, TX) September 2024 - Present

- Researching discriminator-free style transfer for expressive locomotion in humanoids and quadrupeds using RL in Isaaclab to deploy on hardware.
- Developing a custom retargeting algorithm for teleoperation of a robotic arm and hand.
- Developing rudimentary whole-body locomanipulation for the Unitree G1 in IsaacLab.
- Helped develop embedded systems, including CAN hardware and firmware, a ROS2 framework, and an impedance controller, for a novel robotic hand to be controlled via imitation learning.
- Tested and implemented a 6-axis Force/Torque transducer with custom Apptronik amplifier boards in the limbs of humanoids for accurate multi-contact planning using wrench responses.

Lockheed Martin Aeronautics, *Systems Engineering Intern* (Fort Worth, TX) May 2025 – August 2025

- Developed and owned a Python app for automation of sustainment engineering interpretation of F-35 flight data across all Mission System subsystems that improved process workflow by 92%.
- Created, maintained, and documented a codebase with Git, combining a GUI, automated database interaction (SQL), and automated data visualization.

Pike Robotics, *Electrical Engineer* (Austin, TX) September 2023 - September 2024

- Worked independently to design and validate an autonomous embedded system that manages an umbilical line and communicates with a host robot in an oil storage tank environment.
- Designed and implemented the main power distribution PCB at 250W, ensuring UL60950-1 general electrical safety standard under UL 1203 C1D1 explosion-proof environment conditions.

PROJECT EXPERIENCE

Tron Handheld Arcade Console (Austin, TX) Jan 2024 - May 2024

- Recreated the classic Tron arcade bike mini-game on a pair of TI MSPM0 microcontrollers.
- Developed software frontend and backend with hardware drivers in C++ for all devices.
- Implemented LOS wireless infrared UART between two consoles for data and synchronization.

First Tech Challenge 2022/3 - PowerPlay, *Leader* (Houston, TX) August 2022 - May 2023

- Created a codebase in Android Studio to control and interface the subsystems of the robot.
- Led systems integration, coding, and control circuitry, working with the team to design the robot.

Indoor Positioning System, *Individual* (Houston, TX) January 2022 - December 2022

- Developed a 2-dimensional indoor positioning system that uses computer vision to analyze anchor points within rooms and calculate an object's location for delivery robot positioning.

SKILLS AND ACCOMPLISHMENTS

- **Software**: Python, C/C++, SQL, Git, Java, MATLAB, Mathematica, ARM Cortex-M4F
- **Software Packages:** PyTorch, Isaaclab, ROS 2, OpenCV, TensorFlow
- **Hardware**: KiCad, Solidworks, PCB Design, UART, CANbus, SPI, I2C, EtherCAT, LTSpice
- **Languages**: Native English, Native German, Native Russian, Advanced Spanish
- **Awards/Clearances:** DOD Security Clearance (S), Dr. Beer Endowment for Student Excellence